

A SPECULATIVE THEORETICAL PHYSICS PAPER

The Dimensional Speed-Limit Hypothesis

From Geometric Recursion to an Isotropy-Normalized $4 + 1$ Toy Model

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Abstract

This paper documents the development of the Dimensional Speed-Limit Hypothesis (DSLH), a speculative kinematic model in which the maximum causal or propagation speed depends on the number of physically accessible spatial dimensions, while the number of time dimensions remains fixed at one. The construction begins from a Pythagorean recursion and the empirical baseline $C_3 = c$, where C_N denotes the proposed limiting speed in a space with N accessible spatial dimensions. A dimensional-consistency correction separates the newly added coordinate K_N from its velocity contribution κ_N , yielding the central postulate $C_N^2 = C_{N-1}^2 + \kappa_N^2$.

The paper then formulates a one-time-dimensional interval, an effective scalar-wave toy model, and a compact $4 + 1$ realization with characteristic scale $E_4 = \hbar c/R_4$. An energy-dependent accessibility factor is introduced as a phenomenological ansatz. Finally, spatial isotropy is used to normalize the per-axis contribution relative to the observed three-dimensional limit. Under that normalization, a fully accessible additional spatial axis contributes $\kappa^2 = c^2/3$, giving $C_N = c\sqrt{(N/3)}$ in the fully accessible, equal-axis limit, and $C_{4,\max} = 2c/\sqrt{3}$.

The paper explicitly separates standard mathematical and physical ingredients from the novel DSLH postulate. It does not claim experimental evidence for dimension-dependent superluminal propagation; rather, it presents a coherent toy framework, identifies its unresolved dynamical and causal requirements, and states how it could in principle be constrained or falsified.

Keywords: *extra dimensions · causal structure · speed limit · Kaluza–Klein compactification · Lorentz symmetry · effective field theory · speculative kinematics*

1 Introduction

Special relativity assigns a universal invariant speed c to local causal propagation in Minkowski spacetime. Higher-dimensional theories enlarge the number of spatial coordinates but, in their conventional form, do not imply that the invariant speed increases merely because an extra spatial dimension exists. The present work studies a deliberately different hypothesis.

The motivating geometric observation is elementary. If an orthogonal coordinate is added to a Euclidean vector, the squared norm acquires an additional positive term. DSLH asks whether an analogous recursive structure could characterize a maximum kinematic magnitude when a new spatial dimension becomes physically accessible. The proposal keeps exactly one time dimension throughout. Thus the hierarchy considered here is $3 + 1$, $4 + 1$, $5 + 1$, and so on — never $3 + 2$ or another multi-time construction.

The purpose of this paper is not to present the hypothesis as established physics. It documents the reasoning process: the original geometric recursion, the dimensional correction needed to make it physically meaningful, the construction of an effective propagation model, the introduction of a compact fourth dimension, the attempted normalization of the new contribution, and the correction produced by imposing equal per-axis spatial strength.

2 Notation and interpretation

The notation keeps coordinates, velocities, and limiting quantities distinct. This is essential because a spatial coordinate has dimensions of length, whereas a speed has dimensions of length divided by time.

Table 1. Core notation used throughout the paper.

Symbol	Meaning
N	Number of accessible spatial dimensions.
t	The single time coordinate, retained for every N .
c	Observed invariant speed in ordinary $3 + 1$ special relativity.
C_N	Proposed maximum speed when N spatial dimensions are accessible.
K_N	Newly added N th spatial coordinate.
$v_{KN} = dK_N/dt$	Actual velocity component along KN .
κ_N	Proposed maximum kinematic contribution associated with KN .
A_N	Accessibility factor, $0 \leq A_N \leq 1$.
R_N	Characteristic compactification radius.
E_N	Characteristic energy scale for resolving the dimension.
β_N, ξ_N	Dimensionless phenomenological couplings.

The use of κ_N is intentional. The original geometric idea used KN for the newly added dimension. Writing the speed contribution with the same symbol would mix a coordinate with a velocity. KN is therefore reserved for the coordinate, and κ_N for its proposed limiting velocity contribution:

$$|dK_N/dt| \leq \kappa_N \quad (1)$$

3 Stage I: the geometric recursion

The starting point is

$$C_3 = c \quad (2)$$

The central DSLH postulate is

$$C_{N^2} = C_{N-1^2} + \kappa_{N^2}, \quad N \geq 4 \quad (3)$$

Iteration gives

$$C_{N^2} = c^2 + \sum_{j=4}^N \kappa_j^2 \quad (4)$$

so that

$$C_{4^2} = c^2 + \kappa_4^2 \quad (5)$$

$$C_{5^2} = c^2 + \kappa_4^2 + \kappa_5^2 \quad (6)$$

Equation (3) is the defining new postulate. Pythagorean addition of orthogonal vector components is standard mathematics; interpreting a newly accessible spatial dimension as adding a new contribution to the fundamental causal-speed norm is not a consequence of ordinary special relativity.

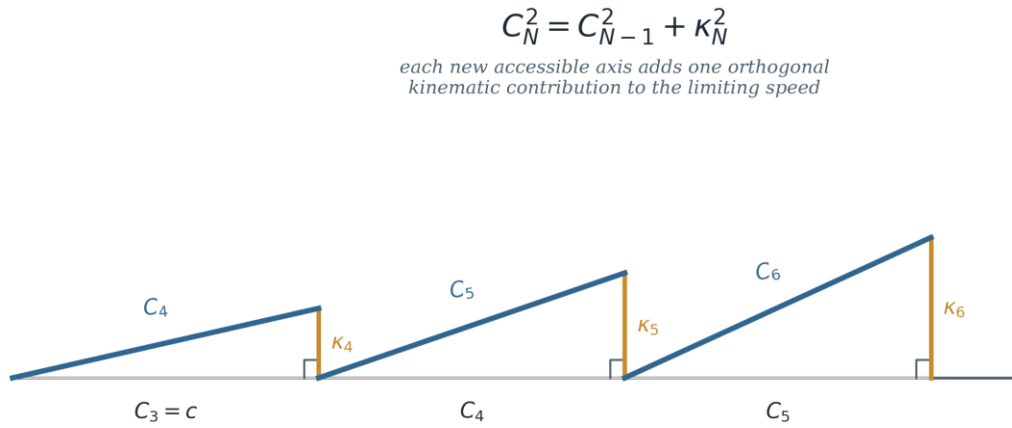


Figure 1. The DSLH recursion read as a sequence of right triangles: each newly accessible spatial axis contributes an orthogonal kinematic term κ_N , so the limiting speed grows as the hypotenuse of an expanding construction.

4 Stage II: exactly one time dimension

DSLH assumes

$$N_{time} = 1 \text{ for every } N \quad (7)$$

A minimal flat interval is

$$ds_{N^2} = \sum_{i=1}^N dx_i^2 - C_{N^2} dt^2 \quad (8)$$

For the observed 3 + 1 regime,

$$ds_{3^2} = dx^2 + dy^2 + dz^2 - c^2 dt^2 \quad (9)$$

and, writing $w \equiv K_4$,

$$ds_{4^2} = dx^2 + dy^2 + dz^2 + dw^2 - C_{4^2} dt^2 \quad (10)$$

No imaginary time coordinate is required; the Lorentzian sign distinguishes the temporal direction. Equation (8), however, does not by itself establish a new relativity principle.

For a null path,

$$\sum_{i=1}^N (dx_i/dt)^2 = C_{N^2} \quad (11)$$

The non-standard content of DSLH is not merely an additional velocity component; it is the claim that the radius of the limiting velocity surface itself may increase.

5 Stage III: accessibility and an effective propagation model

Introduce an accessibility factor

$$0 \leq A_j \leq 1 \quad (12)$$

where $A_j = 0$ denotes an inaccessible extra dimension and $A_j = 1$ a fully accessible one. Define

$$\Gamma_N = 1 + \sum_{j=4}^N \beta_j A_j \quad (13)$$

and postulate

$$C_{N^2} = c^2 \Gamma_N \quad (14)$$

If

$$\kappa_j^2 \equiv c^2 \beta_j A_j \quad (15)$$

then the central recursion of Equation (3) is recovered.

Let ϕ be a massless disturbance with effective action

$$S_N = \frac{1}{2} \int dt d^N x \left[(1/c^2)(\partial\phi/\partial t)^2 - \Gamma_N |\nabla_N \phi|^2 \right] \quad (16)$$

For constant Γ_N , variation gives

$$\partial^2 \phi / \partial t^2 - c^2 \Gamma_N \nabla_N^2 \phi = 0 \quad (17)$$

or, equivalently,

$$\partial^2 \phi / \partial t^2 - C_{N^2} \nabla_N^2 \phi = 0 \quad (18)$$

For a plane wave $\phi \propto \exp[i(\mathbf{q} \cdot \mathbf{x} - \omega t)]$, the dispersion relation is

$$\omega^2 = C_{N^2} |\mathbf{q}|^2 \quad (20)$$

When CN is independent of $|q|$ over the regime considered, the group speed is CN. This is an effective representation, not a fundamental derivation.

6 Stage IV: a compact fourth spatial dimension

Let $w \equiv K_4$ and impose the periodic identification

$$w \sim w + 2\pi R_4 \quad (21)$$

For a compact spatial coordinate, momentum is quantized in the Kaluza–Klein form

$$p_4 = n\hbar / R_4, \quad n = 0, 1, 2, \dots \quad (22)$$

and the first non-zero mode introduces the characteristic scale

$$E_4 = \hbar c / R_4 \quad (23)$$

Compactification supplies this scale; it does not by itself imply a larger causal speed.

7 Stage V: a threshold-motivated accessibility law

An early version used a smooth exponential activation function. A less arbitrary dimensionless form can be motivated from

$$E^2 = p^2 c^2 + E_4^2 \quad (24)$$

which gives

$$p^2 c^2 / E^2 = 1 - (E_4 / E)^2 \quad (25)$$

This motivates, but does not derive, the phenomenological ansatz

$$\begin{aligned} A_4(E) &= 0 \quad \text{for } E < E_4 \\ A_4(E) &= 1 - (E_4/E)^2 \quad \text{for } E \geq E_4 \end{aligned} \quad (26)$$

With dimensionless coupling ξ_4 ,

$$\kappa_4^2(E) = \xi_4^2 c^2 A_4(E) \quad (27)$$

so that

$$C_4(E) = c \sqrt{1 + \xi_4^2 [1 - (E_4/E)^2]}, \quad E \geq E_4 \quad (28)$$

Using Equation (23),

$$C_4(E, R_4) = c \sqrt{1 + \xi_4^2 [1 - (\hbar c / E R_4)^2]} \quad (29)$$

8 Stage VI: the original equal-extra-dimension trial

The first normalization trial set $\xi_4 = 1$. Then

$$C_4(E) = c \sqrt{2 - (E_4/E)^2} \quad (30)$$

with

$$C_{4,max} = \sqrt{2} \cdot c \quad (31)$$

Applying the same full c^2 contribution to every additional dimension gives

$$C_N = c \sqrt{(N - 2)} \quad (32)$$

This reproduces the original geometric intuition, but it contains a normalization problem: it assigns an entire contribution c^2 to each new axis, while the three observed spatial axes collectively correspond to only $C_3^2 = c^2$.

9 Stage VII: isotropy normalization and the key correction

Assume that, when spatial dimensions are locally equivalent and fully accessible, each independent spatial axis contributes the same squared kinematic amount u^2 . In the observed three-dimensional spatial sector,

$$C_3^2 = 3 u^2 \quad (33)$$

Using $C_3 = c$ gives

$$u = c / \sqrt{3} \quad (34)$$

Therefore a fourth axis with the same local per-axis normalization contributes

$$\kappa_4^2 = c^2/3 \quad (35)$$

when fully accessible, rather than $\kappa_4^2 = c^2$. For partial accessibility,

$$\kappa_4^2(E) = (c^2/3) A_4(E) \quad (36)$$

which corresponds to $\xi_4 = 1/\sqrt{3}$. Thus

$$C_4(E) = c \sqrt{(1 + (1/3) A_4(E))} \quad (38)$$

and, explicitly,

$$\begin{aligned} C_4(E) &= c \quad \text{for } E < E_4 \\ C_4(E) &= c \sqrt{(1 + \frac{1}{3} [1 - (E_4/E)^2])} \quad \text{for } E \geq E_4 \end{aligned} \quad (39)$$

In terms of the compactification radius,

$$C_4(E, R_4) = c \sqrt{(1 + (1/3) [1 - (\hbar c / E R_4)^2])} \quad (40)$$

for $ER_4 \geq \hbar c$. The high-energy limit is

$$C_{4,max} = (2/\sqrt{3}) c \approx 1.155 c \quad (41)$$

The symmetry-normalized analysis therefore does not preserve the original $\sqrt{2} c$ trial value for four spatial dimensions.

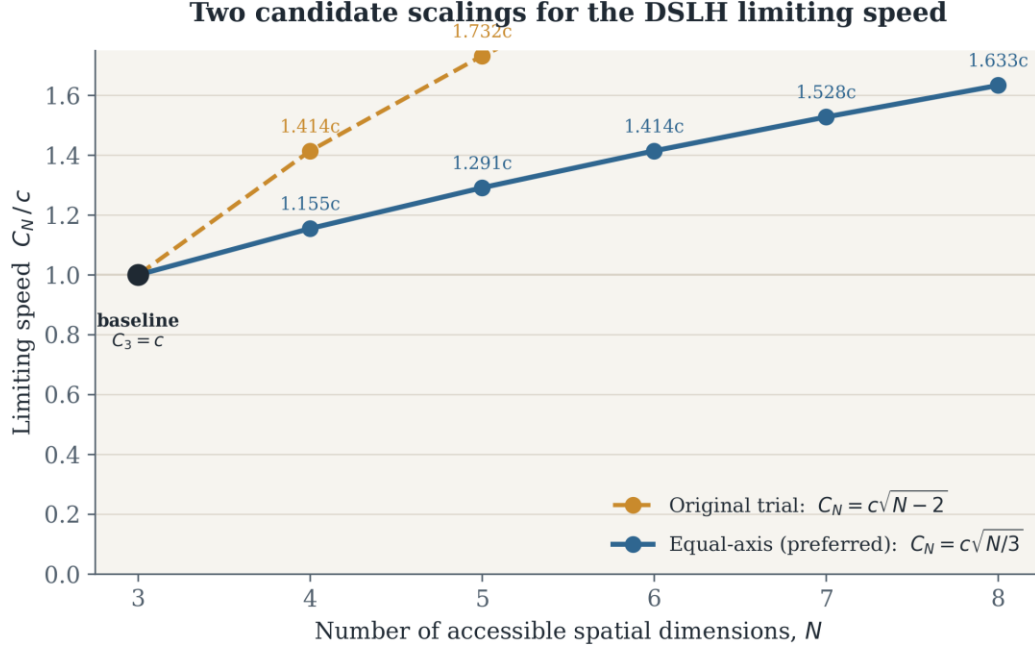


Figure 2. The original trial $C_N = c\sqrt{N-2}$ versus the isotropy-normalized (preferred) scaling $C_N = c\sqrt{N/3}$. Equal per-axis weighting produces a noticeably shallower growth of the limiting speed with dimension count.

10 Stage VIII: the fully accessible N-dimensional limit

If every resolved spatial axis has the same fully accessible per-axis contribution $u^2 = c^2/3$, then

$$C_{N^2} = N u^2 = (N/3) c^2 \quad (42)$$

and hence

$$C_N = c \sqrt{N/3} \quad (43)$$

The first values are

$$C_3 = c \quad (44)$$

$$C_4 = (2/\sqrt{3}) c \approx 1.155 c \quad (45)$$

$$C_5 = \sqrt{(5/3)} c \approx 1.291 c \quad (46)$$

$$C_6 = \sqrt{2} \cdot c \approx 1.414 c \quad (47)$$

For multiple partially accessible extra dimensions,

$$C_{N^2}(E) = c^2 [1 + (1/3) \sum_{j=q}^N A_j(E)] \quad (48)$$

A compact-dimension phenomenological family is

$$E_j = \hbar c / R_j \quad (49)$$

$$\begin{aligned} A_j(E) &= 0 \quad \text{for } E < E_j \\ A_j(E) &= 1 - (E_j/E)^2 \quad \text{for } E \geq E_j \end{aligned} \quad (50)$$

11 Interpretation of the isotropy argument

The normalization above is conditional. If the fourth dimension is compact while the observed three spatial dimensions are effectively non-compact, the global geometry is not fully rotationally invariant under $SO(4)$. The argument should therefore be read as a local equal-axis normalization in a regime where the extra coordinate is physically resolved — not as a theorem that compactification itself enforces $\xi_4 = 1/\sqrt{3}$.

The development therefore encountered two distinct normalization choices, summarized below. The equal-axis choice is preferred within this paper because its normalization is tied directly to $C_3 = c$ and equal per-axis squared contributions.

Table 2. Normalization choices encountered during development.

Model	New-axis contribution	Fully accessible CN
Original trial	$\kappa j^2 = c^2$	$C = c \sqrt{(N - 2)}$
Equal-axis normalization (preferred)	$\kappa j^2 = c^2/3$	$C = c \sqrt{(N/3)}$

12 Energy–momentum relation and self-consistency

For the massless scalar toy model with constant CN,

$$E^2 = C_{N^2} |p|^2 \quad (51)$$

A provisional massive analogue would be

$$E^2 = m^2 C_{N^4} + C_{N^2} |p|^2 \quad (52)$$

Equation (52) cannot be treated as derived when $CN = CN(E)$. Energy then appears on both sides, and the physical group velocity must be obtained from the complete implicit relation,

$$v_g = dE / d|p| \quad (53)$$

A fundamental theory must derive its dispersion relation and accessibility dynamics together.

13 Lorentz symmetry and causality

A single time dimension avoids the separate complications of multiple timelike coordinates, but it does not prove causal consistency. In ordinary special relativity, superluminal signalling combined with universal Lorentz transformations can generate frame-ordering problems. Precision experiments also constrain Lorentz violation strongly.

If DSLH permits $CN > c$ in a higher-dimensional sector, a complete realization must supply a modified symmetry group, preferred foliation, multi-metric structure, restricted sector coupling, or another consistent causal mechanism. A constant replacement $c \rightarrow CN$ in a single universal metric may also be removable by rescaling time, unless the normalization of t is fixed by an independently observable sector. This remains a central unresolved issue.

14 Relation to established extra-dimensional physics

Kaluza–Klein theory, large-extra-dimension models, and warped geometries provide frameworks in which extra spatial dimensions alter spectra or gravity. In standard compactification, momentum along a compact dimension appears as an effective mass contribution to lower-dimensional observers; it does not automatically enlarge the invariant local speed c .

The logical status of the ingredients is therefore:

1. Orthogonal norm addition is standard mathematics.
2. Compact momentum quantization and the scale $\hbar c/R_j$ are standard extra-dimensional ingredients.
3. Interpreting A_j as causal-speed accessibility is phenomenological.
4. The recurrence $CN^2 = CN_{-1}^2 + \kappa N^2$ is the defining DSLH conjecture.
5. The equal-axis result $\kappa^2 = c^2/3$ follows from the DSLH normalization $C_3 = c$ plus equal squared contributions per axis; it is not a prediction of conventional Kaluza–Klein theory.

15 Phenomenology and falsifiability

A scientifically useful DSLH realization must generate observables. Candidate signatures could include threshold-dependent propagation, modified dispersion, missing momentum into extra dimensions, short-distance gravitational deviations, or sector-specific Lorentz-violating effects. Existing collider searches, astrophysical propagation tests, short-range gravity measurements, and precision Lorentz tests constrain many such possibilities.

The model becomes falsifiable once R_j , the accessibility dynamics, and the couplings are sufficiently specified. A realization should be rejected if it predicts already-excluded Lorentz violation, unstable modes, inconsistent causal loops, or extra-dimensional effects incompatible with experiment.

16 Development summary

The conceptual development proceeds as

$$\text{geometric recursion} \rightarrow \text{dimensional correction} \rightarrow \text{accessibility} \rightarrow \text{wave dynamics} \rightarrow \\ \text{compactification scale} \rightarrow \text{isotropy normalization}$$

and, for the concrete $4 + 1$ construction,

$$R_4 \rightarrow E_4 = \hbar c/R_4 \rightarrow A_4(E) \rightarrow \kappa_4(E) \rightarrow C_4(E)$$

The key correction was recognizing that setting the contribution of each new axis equal to c^2 is not equivalent to treating all spatial axes equally. Equal per-axis normalization relative to $C_3 = c$ gives $c/\sqrt{3}$ per fully accessible axis.

17 Open problems

The framework remains incomplete in decisive respects. The accessibility law $A_j(E)$ should be derived dynamically rather than chosen phenomenologically. The effective propagation coefficient should arise from a stable field theory or geometry. A transformation law and globally consistent

causal structure are required. Massive and massless dispersion relations must be derived self-consistently when accessibility depends on energy. Finally, all parameters must be confronted quantitatively with experiment.

A deeper theory may preserve the equal-axis scaling, modify it, or eliminate the dimension-dependent causal-speed interpretation altogether. The present construction makes those questions explicit and mathematically testable.

18 Conclusion

DSLH begins with

$$C_{N^2} = C_{N-1^2} + \kappa_{N^2} \quad (56)$$

with $C_3 = c$ and exactly one time dimension. Separating the spatial coordinate κN from its limiting velocity contribution κN resolves the original dimensional ambiguity. An effective scalar-wave action shows how C_N can function as a propagation speed in a toy model, while compactification supplies the scale $E_j = \hbar c/R_j$ for resolving an extra dimension.

The development also corrects the earliest normalization. If all resolved spatial axes are assigned equal squared kinematic weight and the observed three-axis total is normalized to c^2 , each fully accessible axis contributes $c^2/3$. The corresponding fully accessible hierarchy is

$$C_N = c \sqrt{(N/3)} \quad (57)$$

not $c\sqrt{(N-2)}$. In particular,

$$C_{4,max} = (2/\sqrt{3}) c \quad (58)$$

This result is conditional on the DSLH equal-axis normalization and is not an established law of nature. A viable theory still requires a dynamical derivation, causal consistency, a non-trivial physical meaning for C_N/c , and agreement with precision tests of relativity and extra-dimensional physics.

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Declarations

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Conflict of interest

The author declares no competing interests.

Data availability

No datasets were generated or analysed in this theoretical study.

References

- [1] Appelquist, T., Chodos, A., Freund, P. G. O. (eds.) *Modern Kaluza–Klein Theories*. Addison-Wesley (1987).
- [2] Arkani-Hamed, N., Dimopoulos, S., Dvali, G. The hierarchy problem and new dimensions at a millimeter. *Physics Letters B* 429, 263–272 (1998). doi:10.1016/S0370-2693(98)00466-3
- [3] Einstein, A. Zur Elektrodynamik bewegter Körper. *Annalen der Physik* 322(10), 891–921 (1905). doi:10.1002/andp.19053221004
- [4] Hewett, J. L., Spiropulu, M. Particle physics probes of extra spacetime dimensions. *Annual Review of Nuclear and Particle Science* 52, 397–424 (2002). doi:10.1146/annurev.nucl.52.050102.090706
- [5] Kaluza, T. Zum Unitätsproblem der Physik. *Sitzungsberichte der Preussischen Akademie der Wissenschaften*, 966–972 (1921).
- [6] Klein, O. Quantentheorie und fünfdimensionale Relativitätstheorie. *Zeitschrift für Physik* 37, 895–906 (1926). doi:10.1007/BF01397481
- [7] Kostelecký, V. A., Russell, N. Data tables for Lorentz and CPT violation. *Reviews of Modern Physics* 83, 11–31 (2011). doi:10.1103/RevModPhys.83.11
- [8] Mattingly, D. Modern tests of Lorentz invariance. *Living Reviews in Relativity* 8(5) (2005). doi:10.12942/lrr-2005-5
- [9] Randall, L., Sundrum, R. A large mass hierarchy from a small extra dimension. *Physical Review Letters* 83, 3370–3373 (1999). doi:10.1103/PhysRevLett.83.3370